

Instructor: He Wang

Name: _____

To receive full credit for a problem you must show all necessary work.

1. (10 points) Use the arc length formula to find the length of the curve

$$y = 3(2x - 1)^{3/2}$$

for $1 \leq x \leq 3$.

$$y' = \frac{9}{2} (2x-1)^{\frac{1}{2}} \cdot 2 = 9(2x-1)^{\frac{1}{2}}$$

By arc length formula

$$L = \int_1^3 \sqrt{1 + (y')^2} dx = \int_1^3 \sqrt{1 + 81(2x-1)} dx$$

$$= \int_1^3 \sqrt{162x - 80} dx$$

$$u = 162x - 80$$

$$du = 162 dx$$

$$dx = \frac{1}{162} du$$

$$= \int_{82}^{406} u^{\frac{1}{2}} \frac{1}{162} du$$

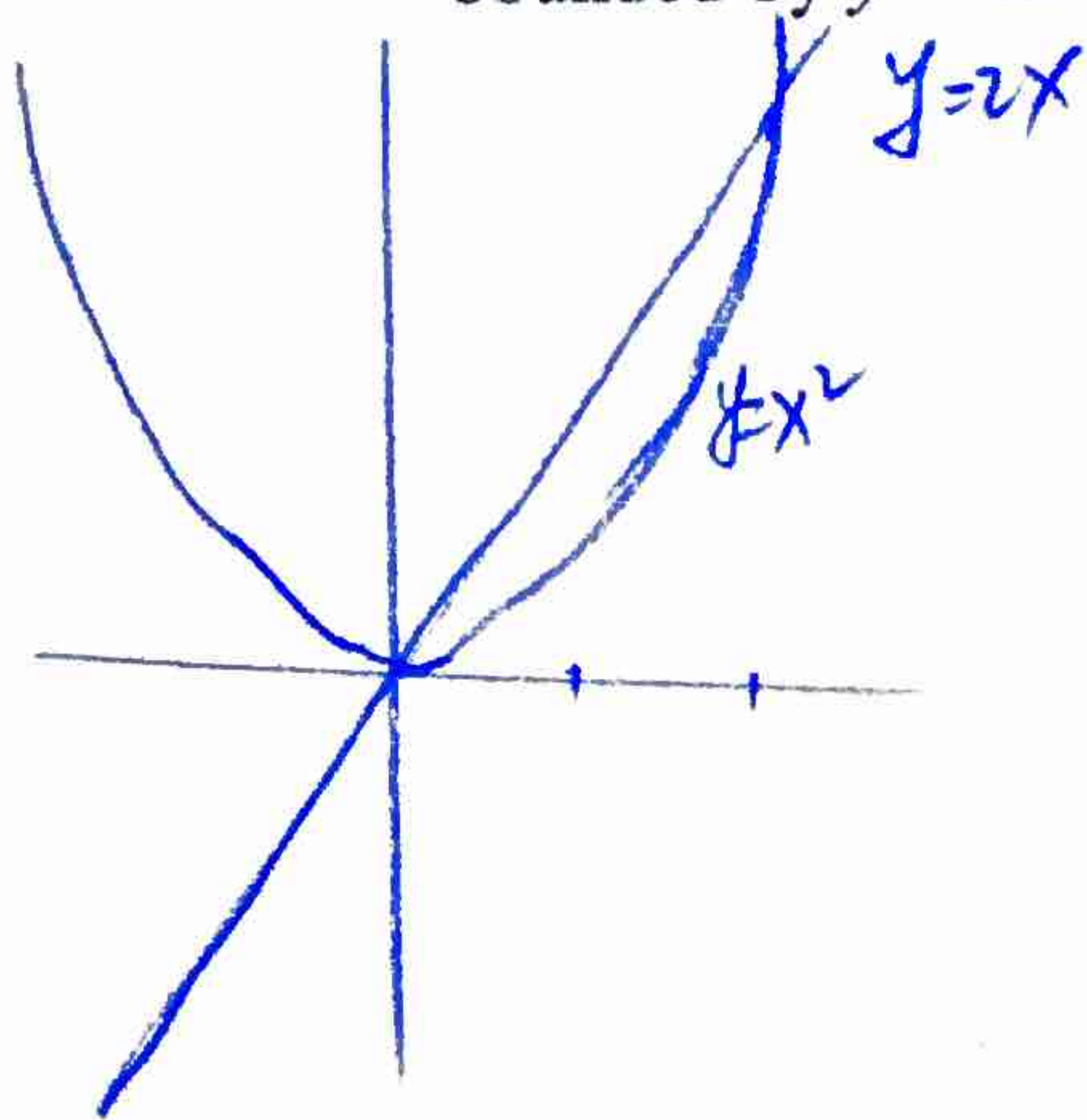
$$u(1) = 82$$

$$u(3) = 406$$

$$= \frac{1}{162} \cdot \frac{2}{3} u^{\frac{3}{2}} \Big|_{82}^{406}$$

$$= \frac{1}{243} (406^{\frac{3}{2}} - 82^{\frac{3}{2}})$$

2. (10 points) Calculate the center of mass (centroid) of a lamina with density $\rho = 6$ and shape R bounded by $y = 2x$ and the parabola $y = x^2$.



① The area of the region R is

$$A = \int_0^2 2x - x^2 dx = \left. x^2 - \frac{x^3}{3} \right|_0^2 = \frac{4}{3}$$

The intersection,

$$2x = x^2$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x=0 \text{ and } x=2$$

$$\textcircled{2} \bar{x} = \frac{1}{A} \int_0^2 x(f(x) - g(x)) dx$$

$$= \frac{3}{4} \int_0^2 x(2x - x^2) dx$$

$$= \frac{3}{4} \int_0^2 (2x^2 - x^3) dx$$

$$= \frac{3}{4} \left(\frac{2}{3} x^3 - \frac{x^4}{4} \right) \Big|_0^2 = \frac{3}{4} \left(\frac{2 \cdot 2^3}{3} - \frac{2^4}{4} \right) = 1$$

$$\textcircled{3} \bar{y} = \frac{1}{2A} \int_0^2 f(x)^2 - g(x)^2 dx$$

$$= \frac{1}{2} \cdot \frac{3}{4} \int_0^2 4x^2 - x^4 dx$$

$$= \frac{3}{8} \left(\frac{4}{3} x^3 - \frac{x^5}{5} \right) \Big|_0^2 = \frac{8}{5}$$

So, the centroid of the lamina is $(1, \frac{8}{5})$

3. (1) (6 points) For what values of r does the function $y = e^{rx}$ a solution for $y'' - 4y = 0$?

$$y' = r e^{rx}$$

$$y'' = r^2 e^{rx}$$

plug into the equation

$$r^2 e^{rx} - 4e^{rx} = 0$$

$$(r^2 - 4) e^{rx} = 0$$

since $e^{rx} \neq 0$, we have

$$r^2 - 4 = 0$$

$$(r+2)(r-2) = 0$$

$$r = 2 \text{ and } r = -2$$

So $y = e^{2x}$ and $y = e^{-2x}$ are
solutions for $y'' - 4y = 0$

(2) (4 points) If r_1 and r_2 are the values of r that you found in part (1), show that every member of the family of functions $y = ae^{r_1 x} + be^{r_2 x}$ is a solution for $y'' - 4y = 0$.

$$y = ae^{2x} + be^{-2x}$$

$$y' = 2ae^{2x} - 2be^{-2x}$$

$$y'' = 4ae^{2x} + 4be^{-2x}$$

$$\text{left side of the equation} = y'' - 4y$$

$$= 4ae^{2x} + 4be^{-2x} - 4(ae^{2x} + be^{-2x})$$

$$= 4ae^{2x} + 4be^{-2x} - 4ae^{2x} - 4be^{-2x}$$

$$= 0$$

4. (10 points) Use Euler's method with step size $h = 0.2$ to solve $\frac{dy}{dx} = 2xy - 1$ with initial value $y(1) = 1$. Find $y(1.6)$.

$$x_0 = 1$$

$$y_0 = 1$$

$$x_1 = x_0 + h = 1.2$$

$$y_1 = y_0 + h(2x_0 y_0 - 1)$$

$$x_2 = x_1 + h = 1.4$$

$$= 1 + 0.2(2 - 1) = 1.2$$

$$x_3 = x_2 + h = 1.6$$

$$y_2 = y_1 + h(2x_1 y_1 - 1)$$

$$= 1.2 + 0.2(2(1.2)(1.2) - 1)$$

$$= 1.576$$

$$y_3 = y_2 + h(2x_2 y_2 - 1)$$

$$= 1.576 + 0.2(2(1.4)(1.576) - 1)$$

$$= 2.25856$$

$$\text{So, } y(1.6) \approx y_3 = 2.25856$$

5. (10 points) Solve the differential equation $\frac{dy}{dx} = 4y \cos x$ with the initial condition $y(0) = 3$.

$$\frac{dy}{y} = \cos x dx$$

$$\int \frac{1}{y} dy = \int \cos x dx$$

$$\ln|y| = \sin x + C$$

$$|y| = e^{\sin x + C} = e^{\sin x} \cdot e^C$$

$$y = \pm e^{\sin x} \cdot e^C$$

Denote $a = \pm e^C$, $y = a \cdot e^{\sin x}$

By initial condition $y(0) = 3$

$$3 = a \cdot e^{\sin 0} = a$$

So, the solution is $y = 3e^{\sin x}$

6. (2 bonus points) Solve the differential equation $(\sec^2 y)x^{-2}y' = \sin(2x^3)$

$$\sec^2 y dy = x^2 \sin 2x^3 dx$$

$$\int \sec^2 y dy = \int x^2 \sin 2x^3 dx$$

$$\tan y = -\frac{1}{6} \cos 2x^3 + C$$